pfSense-Based DNS Enforcement & Traffic Control Solution

---

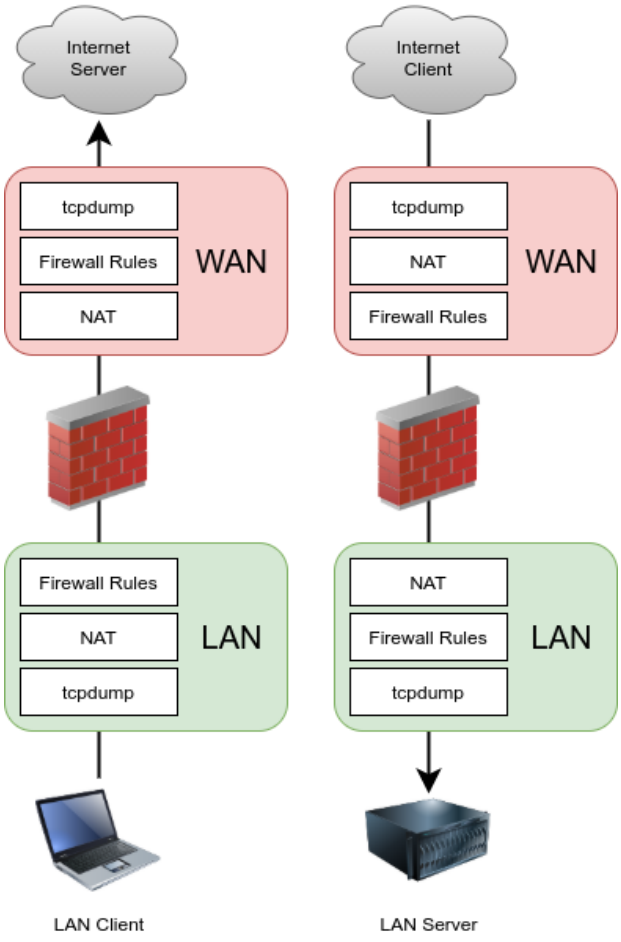
1. Objective

Design and implement a **centralized network control solution** to:

- Enforce DNS usage through firewall
- Block unwanted services (e.g., Windows Update)
- Provide traffic visibility and monitoring
- Reduce unnecessary broadcast/network noise

---

2. High-Level Architecture



```
   +-------------------+
   |   Client Network  |
   | (LAN / OPT1 VLAN) |
   +---------+---------+
             |
             ▼
+------------------------+
|      pfSense FW        |
|------------------------|
| Firewall Rules (L3/L4) |
| DNS Resolver (Unbound) |
| pfBlockerNG (DNSBL)    |
+-----------+------------+
            |
            ▼
+------------------------+
|     Internet / CDN     |
| (Microsoft, Google etc)|
+------------------------+
```

---

3. Components Overview

pfSense Firewall

- Acts as **core gateway**
- Enforces:
  - Routing
  - NAT
  - Firewall policies

DNS Resolver (Unbound)

- Handles DNS queries centrally
- Integrates with pfBlockerNG
- Supports DNS filtering

---

pfBlockerNG (DNSBL)

- DNS-based filtering engine
- Blocks domains using:
  - Sinkhole (VIP: `10.255.255.1`)
- Supports:
  - Threat feeds
  - Custom domain blocking

Monitoring (Logs / ntopng optional)

- Firewall logs for:
  - Traffic visibility
  - Block analysis
- ntopng (future):
  - Application-level monitoring

---

4. Security & Control Design

DNS Enforcement (Core Control)

```
ALLOW → LAN → pfSense → Port 53
BLOCK → LAN → ANY → Port 53
```

✓ Prevents DNS bypass
✓ Forces all queries through pfSense

DNS Sinkhole Filtering

```
Blocked Domain → 10.255.255.1
```

✓ Redirects malicious/unwanted traffic
✓ Prevents external communication

Windows Update Blocking

Domains blocked:

```
windowsupdate.microsoft.com
update.microsoft.com
download.windowsupdate.com
delivery.mp.microsoft.com
fe2.update.microsoft.com
```

✓ Prevents OS updates via internet

---

5. Traffic Behavior Analysis

Observed Patterns

- High UDP traffic (~90%)
- Top source host:
  - `172.16.0.11` (heavy traffic generator)

Identified Protocols

Protocol	Description
UDP 1900	SSDP (UPnP discovery)
UDP 5353	mDNS
UDP 137	NetBIOS

---

6. Optimization Strategy

Reduce Network Noise

```
Block UDP 1900 (SSDP)
Block UDP 5353 (mDNS)
```

✓ Reduces broadcast traffic
✓ Improves network efficiency

Validation Strategy

DNS Validation

```
nslookup windowsupdate.microsoft.com
```

Expected:

```
10.255.255.1 (Blocked)
```

Connectivity Validation

```
ping google.com → SUCCESS
```

✓ Internet unaffected
✓ Only targeted blocking

---

8. Troubleshooting Summary

Issue Encountered

- Package manager failure:

```
libssl.so.30 not found
```

Resolution

```
pkg-static bootstrap -f
pkg-static update -f
```

✓ Restored package system

---

9. Data Flow (Logical)

```
Client
  ↓
DNS Request
  ↓
pfSense (Unbound)
  ↓
pfBlockerNG Check
  ├── Allowed → Internet
  └── Blocked → 10.255.255.1
```

---

10. Limitations

- Windows Update uses CDN → partial bypass possible
- DNS-only blocking (no deep packet inspection)
- No user-level identity control (no AD integration yet)

---

11. Future Enhancements

Monitoring

- Deploy **ntopng** for:
  - Application visibility
  - Bandwidth tracking

Security

- Add **Suricata (IPS)**
- Enable threat detection

Enterprise Integration

- Integrate **Active Directory DNS**
- Deploy **WSUS** for controlled updates

Application Control

- Block:
  - YouTube
  - Streaming services
  - Social media

---

12. Business Outcome

✓ Centralized control
✓ Reduced unwanted traffic
✓ Improved visibility
✓ Enhanced security posture

---

13. Interview Summary (1-Line)

"Designed a pfSense-based DNS enforcement architecture using pfBlockerNG to implement domain-level traffic control, prevent DNS bypass, and optimize network behavior through traffic analysis."

Final Status

Capability	Status
DNS Enforcement	🔥 Implemented
Traffic Control	🔥 Strong
Monitoring	✅ Available
Security	✅ Moderate (expandable)